JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Sequences and Series of Real Numbers (2015–2021)

Category: Real Analysis • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
60	100 M	180 Min	MCQ • MSQ • NAT (2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	33	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	8	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	19	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q55 • ID: JAM_2021_Q55) NAT +2 M Neg: Nil (0)

Q. 55 Define the sequence

$$s_n = \begin{cases} \dfrac{1}{2^n}\displaystyle\sum_{j=0}^{n-2} 2^{2j} & \text{if } n > 0 \text{ is even,} \\[2ex] \dfrac{1}{2^n}\displaystyle\sum_{j=0}^{n-1} 2^{2j} & \text{if } n > 0 \text{ is odd.} \end{cases}$$

Define $\sigma_m = \dfrac{1}{m}\displaystyle\sum_{n=1}^{m} s_n$. The number of limit points of the sequence $\{\sigma_m\}$ is _____.

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Question 2 (JAM 2021 • Q50 • ID: JAM_2021_Q50) NAT +1 M Neg: Nil (0)

Q. 50 The value of

$$\frac{\pi}{2} \lim_{n\to\infty} \cos\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{8}\right)\cdots\cos\left(\frac{\pi}{2^{n+1}}\right)$$

is _____.

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Question 3 (JAM 2021 • Q42 • ID: JAM_2021_Q42) NAT +1 M Neg: Nil (0)

Q. 42 The value of

$$\lim_{n\to\infty} \left(3^n + 5^n + 7^n\right)^{\frac{1}{n}}$$

is _____.

Question 4 (JAM 2021 • Q30 • ID: JAM_2021_Q30) MCQ +2 M Neg: −0.67

Q. 30 Consider the two series

$$\text{I. } \sum_{n=1}^{\infty} \frac{1}{n^{1+(1/n)}} \quad \text{and} \quad \text{II. } \sum_{n=1}^{\infty} \frac{1}{n^{2-n^{1/n}}}.$$

Which one of the following holds?

(A) Both I and II converge.

(B) Both I and II diverge.

(C) I converges and II diverges.

(D) I diverges and II converges.

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Question 5 (JAM 2021 • Q15 • ID: JAM_2021_Q15) MCQ +2 M Neg: −0.67

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Q. 15 Let $f : \mathbb{N} \to \mathbb{N}$ be a bijective map such that

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^2} < +\infty.$$

The number of such bijective maps is

(A) exactly one.
(B) zero.
(C) finite but more than one.
(D) infinite.

Question 6 (JAM 2020 • Q52 • ID: JAM_2020_Q52) NAT +2 M Neg: Nil (0)

Q. 52 Consider the expansion of the function $f(x) = \dfrac{3}{(1-x)(1+2x)}$ in powers of x, that is valid in $|x| < \dfrac{1}{2}$. Then the coefficient of x^4 is ________.

Question 7 (JAM 2020 • Q51 • ID: JAM_2020_Q51) NAT +2 M Neg: Nil (0)

Q. 51 The sum of the series $\dfrac{1}{2(2^2-1)} + \dfrac{1}{3(3^2-1)} + \dfrac{1}{4(4^2-1)} + \cdots$ is ________.

Question 8 (JAM 2020 • Q41 • ID: JAM_2020_Q41) NAT +1 M Neg: Nil (0)

Q. 41 Let $x_n = n^{\frac{1}{n}}$ and $y_n = e^{1-x_n}$, $n \in \mathbb{N}$. Then the value of $\lim\limits_{n\to\infty} y_n$ is ________.

Question 9 (JAM 2020 • Q39 • ID: JAM_2020_Q39) MSQ +2 M Neg: Nil (0)

Q. 39 Let f be a real valued function of a real variable, such that $|f^{(n)}(0)| \leq K$ for all $n \in \mathbb{N}$, where $K > 0$. Which of the following is/are true?

(A) $\left|\dfrac{f^{(n)}(0)}{n!}\right|^{\frac{1}{n}} \to 0$ as $n \to \infty$

(B) $\left|\dfrac{f^{(n)}(0)}{n!}\right|^{\frac{1}{n}} \to \infty$ as $n \to \infty$

(C) $f^{(n)}(x)$ exists for all $x \in \mathbb{R}$ and for all $n \in \mathbb{N}$

(D) The series $\sum\limits_{n=1}^{\infty} \dfrac{f^{(n)}(0)}{(n-1)!}$ is absolutely convergent

Question 10 (JAM 2020 • Q37 • ID: JAM_2020_Q37) MSQ +2 M Neg: Nil (0)

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Q. 37 Let $a = \lim\limits_{n\to\infty} \left(\dfrac{1}{n^2} + \dfrac{2}{n^2} + \cdots + \dfrac{(n-1)}{n^2}\right)$ and $b = \lim\limits_{n\to\infty} \left(\dfrac{1}{n+1} + \dfrac{1}{n+2} + \cdots + \dfrac{1}{n+n}\right)$.
Which of the following is/are true?

(A) $a > b$

(B) $a < b$

(C) $ab = \ln\sqrt{2}$

(D) $\dfrac{a}{b} = \ln\sqrt{2}$

Question 11 (JAM 2020 • Q32 • ID: JAM_2020_Q32) MSQ +2 M Neg: Nil (0)

Q. 32 If $s_n = \dfrac{(-1)^n}{2^n + 3}$ and $t_n = \dfrac{(-1)^n}{4n-1}$, $n = 0, 1, 2, ...$, then

(A) $\sum\limits_{n=0}^{\infty} s_n$ is absolutely convergent

(B) $\sum\limits_{n=0}^{\infty} t_n$ is absolutely convergent

(C) $\sum\limits_{n=0}^{\infty} s_n$ is conditionally convergent

(D) $\sum\limits_{n=0}^{\infty} t_n$ is conditionally convergent

Question 12 (JAM 2020 • Q13 • ID: JAM_2020_Q13) MCQ +2 M Neg: −0.67

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Q. 13 Suppose that S is the sum of a convergent series $\sum_{n=1}^{\infty} a_n$. Define $t_n = a_n + a_{n+1} + a_{n+2}$. Then the series $\sum_{n=1}^{\infty} t_n$

(A) diverges

(B) converges to $3S - a_1 - a_2$

(C) converges to $3S - a_1 - 2a_2$

(D) converges to $3S - 2a_1 - a_2$

Question 13 (JAM 2020 • Q12 • ID: JAM_2020_Q12) MCQ +2 M Neg: −0.67

Q. 12 Define $s_1 = \alpha > 0$ and $s_{n+1} = \sqrt{\dfrac{1+s_n^2}{1+\alpha}}$, $n \geq 1$. Which of the following is true?

(A) If $s_n^2 < \dfrac{1}{\alpha}$, then $\{s_n\}$ is monotonically increasing and $\lim_{n\to\infty} s_n = \dfrac{1}{\sqrt{\alpha}}$

(B) If $s_n^2 < \dfrac{1}{\alpha}$, then $\{s_n\}$ is monotonically decreasing and $\lim_{n\to\infty} s_n = \dfrac{1}{\alpha}$

(C) If $s_n^2 > \dfrac{1}{\alpha}$, then $\{s_n\}$ is monotonically increasing and $\lim_{n\to\infty} s_n = \dfrac{1}{\sqrt{\alpha}}$

(D) If $s_n^2 > \dfrac{1}{\alpha}$, then $\{s_n\}$ is monotonically decreasing and $\lim_{n\to\infty} s_n = \dfrac{1}{\alpha}$

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Question 14 (JAM 2020 • Q11 • ID: JAM_2020_Q11) **MCQ** **+2 M** **Neg: −0.67**

Q. 11 Let $\{a_n\}$ be a sequence of positive real numbers. Suppose that $l = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$. Which of the following is true?

(A) If $l = 1$, then $\lim_{n\to\infty} a_n = 1$

(B) If $l = 1$, then $\lim_{n\to\infty} a_n = 0$

(C) If $l < 1$, then $\lim_{n\to\infty} a_n = 1$

(D) If $l < 1$, then $\lim_{n\to\infty} a_n = 0$

Question 15 (JAM 2020 • Q9 • ID: JAM_2020_Q9) **MCQ** **+1 M** **Neg: −0.33**

Q. 9 The radius of convergence of the power series

$$\sum_{n=1}^{\infty} \left(\frac{n+2}{n}\right)^{n^2} x^n$$

is

(A) e^2

(B) $\frac{1}{\sqrt{e}}$

(C) $\frac{1}{e}$

(D) $\frac{1}{e^2}$

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Question 16 (JAM 2020 • Q1 • ID: JAM_2020_Q1) **MCQ** **+1 M** **Neg: −0.33**

Q. 1 Let $s_n = 1 + \frac{(-1)^n}{n}, n \in \mathbb{N}$. Then the sequence $\{s_n\}$ is

(A) monotonically increasing and is convergent to 1

(B) monotonically decreasing and is convergent to 1

(C) neither monotonically increasing nor monotonically decreasing but is convergent to 1

(D) divergent

Question 17 (JAM 2019 • Q52 • ID: JAM_2019_Q52) NAT +2 M Neg: Nil (0)

Q.52 The greatest lower bound of the set

$$\{(e^n + 2^n)^{\frac{1}{n}} : n \in \mathbb{N}\},$$

(round off to 2 decimal places) is ______________

Question 18 (JAM 2019 • Q40 • ID: JAM_2019_Q40) MSQ +2 M Neg: Nil (0)

Q.40 Let $\{a_n\}$ be the sequence of real numbers such that

$$a_1 = 1 \text{ and } a_{n+1} = a_n + a_n^2 \text{ for all } n \geq 1.$$

Then

(A) $a_4 = a_1(1 + a_1)(1 + a_2)(1 + a_3)$

(B) $\lim\limits_{n\to\infty} \frac{1}{a_n} = 0$

(C) $\lim\limits_{n\to\infty} \frac{1}{a_n} = 1$

(D) $\lim\limits_{n\to\infty} a_n = 0$

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Question 19 (JAM 2019 • Q36 • ID: JAM_2019_Q36) MSQ +2 M Neg: Nil (0)

Q.36 Let $\{a_n\}$ be the sequence given by

$$a_n = \max\left\{\sin\left(\frac{n\pi}{3}\right), \cos\left(\frac{n\pi}{3}\right)\right\}, \quad n \geq 1.$$

Then which of the following statements is/are TRUE about the subsequences $\{a_{6n-1}\}$ and $\{a_{6n+4}\}$?

(A) Both the subsequences are convergent
(B) Only one of the subsequences is convergent
(C) $\{a_{6n-1}\}$ converges to $-\frac{1}{2}$
(D) $\{a_{6n+4}\}$ converges to $\frac{1}{2}$

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Question 20 (JAM 2019 • Q30 • ID: JAM_2019_Q30) MCQ +2 M Neg: −0.67

Q.30 Let $\{a_n\}$ be a sequence of positive real numbers such that

$$a_1 = 1,\ a_{n+1}^2 - 2a_n a_{n+1} - a_n = 0 \text{ for all } n \geq 1.$$

Then the sum of the series $\sum_{n=1}^{\infty} \frac{a_n}{3^n}$ lies in the interval

(A) $(1, 2]$ (B) $(2, 3]$ (C) $(3, 4]$ (D) $(4, 5]$

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Question 21 (JAM 2019 • Q28 • ID: JAM_2019_Q28) MCQ +2 M Neg: −0.67

Q.28 Let $\{a_n\}$ be a sequence of positive real numbers. The series $\sum_{n=1}^{\infty} a_n$ converges if the series

(A) $\sum_{n=1}^{\infty} a_n^2$ converges
(B) $\sum_{n=1}^{\infty} \frac{a_n}{2^n}$ converges
(C) $\sum_{n=1}^{\infty} \frac{a_{n+1}}{a_n}$ converges
(D) $\sum_{n=1}^{\infty} \frac{a_n}{a_{n+1}}$ converges

Question 22 (JAM 2019 • Q19 • ID: JAM_2019_Q19) MCQ +2 M Neg: −0.67

Q.19 Which one of the following series is divergent?

(A) $\sum_{n=1}^{\infty} \frac{1}{n} \sin^2 \frac{1}{n}$

(B) $\sum_{n=1}^{\infty} \frac{1}{n} \log n$

(C) $\sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{1}{n}$

(D) $\sum_{n=1}^{\infty} \frac{1}{n} \tan \frac{1}{n}$

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Question 23 (JAM 2019 • Q15 • ID: JAM_2019_Q15) MCQ +2 M Neg: −0.67

Q.15 For $-1 < x < 1$, the sum of the power series $1 + \sum_{n=2}^{\infty} (-1)^{n-1} n^2 x^{n-1}$ is

(A) $\frac{1-x}{(1+x)^3}$

(B) $\frac{1+x^2}{(1+x)^4}$

(C) $\frac{1-x}{(1+x)^2}$

(D) $\frac{1+x^2}{(1+x)^3}$

Question 24 (JAM 2019 • Q4 • ID: JAM_2019_Q4) MCQ +1 M Neg: −0.33

Q.4 Let $\{a_n\}_{n=0}^{\infty}$ and $\{b_n\}_{n=0}^{\infty}$ be sequences of positive real numbers such that $na_n < b_n < n^2 a_n$ for all $n \geq 2$. If the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n x^n$ is 4, then the power series $\sum_{n=0}^{\infty} b_n x^n$

(A) converges for all x with $|x| < 2$

(B) converges for all x with $|x| > 2$

(C) does not converge for any x with $|x| > 2$

(D) does not converge for any x with $|x| < 2$

Question 25 (JAM 2019 • Q1 • ID: JAM_2019_Q1) MCQ +1 M Neg: −0.33

Q. 1 – Q.10 carry one mark each.

Q.1 Let $a_1 = b_1 = 0$, and for each $n \geq 2$, let a_n and b_n be real numbers given by

$$a_n = \sum_{m=2}^{n} \frac{(-1)^m m}{(\log(m))^m} \text{ and } b_n = \sum_{m=2}^{n} \frac{1}{(\log(m))^m}.$$

Then which one of the following is TRUE about the sequences $\{a_n\}$ and $\{b_n\}$?

(A) Both $\{a_n\}$ and $\{b_n\}$ are divergent
(B) $\{a_n\}$ is convergent and $\{b_n\}$ is divergent
(C) $\{a_n\}$ is divergent and $\{b_n\}$ is convergent
(D) Both $\{a_n\}$ and $\{b_n\}$ are convergent

Question 26 (JAM 2018 • Q51 • ID: JAM_2018_Q51) NAT +2 M Neg: Nil (0)

Q. 51 – Q. 60 carry two marks each.

Q.51 Let $a_k = (-1)^{k-1}$, $s_n = a_1 + a_2 + \cdots + a_n$ and $\sigma_n = (s_1 + s_2 + \cdots + s_n)/n$, where $k, n \in \mathbb{N}$.

Then $\lim_{n\to\infty} \sigma_n$ is ______________ (correct up to one decimal place).

Question 27 (JAM 2018 • Q47 • ID: JAM_2018_Q47) NAT +1 M Neg: Nil (0)

Q.47 Let $a_n = \frac{(1+(-1)^n)}{2^n} + \frac{(1+(-1)^{n-1})}{3^n}$. Then the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ about $x = 0$ is __________

Question 28 (JAM 2018 • Q43 • ID: JAM_2018_Q43) NAT +1 M Neg: Nil (0)

Q.43 Let $f(x) = \sum_{n=0}^{\infty} (-1)^n x(x-1)^n$ for $0 < x < 2$. Then the value of $f\left(\frac{\pi}{4}\right)$ is ____________

Question 29 (JAM 2018 • Q14 • ID: JAM_2018_Q14) MCQ +2 M Neg: −0.67

Q.14 Let $a_n = \frac{(-1)^n}{\sqrt{1+n}}$ and let $c_n = \sum_{k=0}^{n} a_{n-k} a_k$, where $n \in \mathbb{N} \cup \{0\}$. Then which one of the following is TRUE?

(A) Both $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=1}^{\infty} c_n$ are convergent

(B) $\sum_{n=0}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} c_n$ is not convergent

(C) $\sum_{n=1}^{\infty} c_n$ is convergent but $\sum_{n=0}^{\infty} a_n$ is not convergent

(D) Neither $\sum_{n=0}^{\infty} a_n$ nor $\sum_{n=1}^{\infty} c_n$ is convergent

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Question 30 (JAM 2018 • Q13 • ID: JAM_2018_Q13) MCQ +2 M Neg: −0.67

Q.13 Let $a_n = n + \frac{1}{n}$, $n \in \mathbb{N}$. Then the sum of the series $\sum_{n=1}^{\infty}(-1)^{n+1} \frac{a_{n+1}}{n!}$ is

(A) $e^{-1} - 1$ (B) e^{-1} (C) $1 - e^{-1}$ (D) $1 + e^{-1}$

Question 31 (JAM 2018 • Q12 • ID: JAM_2018_Q12) MCQ +2 M Neg: −0.67

Q.12 Let $a, b, c \in \mathbb{R}$. Which of the following values of a, b, c do NOT result in the convergence of the series

$$\sum_{n=3}^{\infty} \frac{a^n}{n^b (\log_e n)^c} \quad ?$$

(A) $|a| < 1,\ b \in \mathbb{R}, c \in \mathbb{R}$ (B) $a = 1,\ b > 1,\ c \in \mathbb{R}$

(C) $a = 1, b \geq 0, c < 1$ (D) $a = -1,\ b \geq 0, c > 0$

Question 32 (JAM 2018 • Q11 • ID: JAM_2018_Q11) MCQ +2 M Neg: −0.67

Q. 11 – Q. 30 carry two marks each.

Q.11 Let $a_n = \begin{cases} 2 + \frac{(-1)^{\frac{n-1}{2}}}{n}, & if\ n\ is\ odd \\ 1 + \frac{1}{2^n}, & if\ n\ is\ even \end{cases}$, $n \in \mathbb{N}$.

Then which one of the following is TRUE?

(A) $\sup\{a_n \mid n \in \mathbb{N}\} = 3$ and $\inf\{a_n \mid n \in \mathbb{N}\} = 1$

(B) $\liminf (a_n) = \limsup (a_n) = \frac{3}{2}$

(C) $\sup\{a_n \mid n \in \mathbb{N}\} = 2$ and $\inf\{a_n \mid n \in \mathbb{N}\} = 1$

(D) $\liminf (a_n) = 1$ and $\limsup (a_n) = 3$

Question 33 (JAM 2018 • Q10 • ID: JAM_2018_Q10) MCQ +1 M Neg: −0.33

Q.10 Let $s_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$ for $n \in \mathbb{N}$. Then which one of the following is TRUE for the sequence $\{s_n\}_{n=1}^{\infty}$

(A) $\{s_n\}_{n=1}^{\infty}$ converges in $\mathbb{Q}$

(B) $\{s_n\}_{n=1}^{\infty}$ is a Cauchy sequence but does not converge in $\mathbb{Q}$

(C) the subsequence $\{s_{k^n}\}_{n=1}^{\infty}$ is convergent in $\mathbb{R}$, only when k is even natural number

(D) $\{s_n\}_{n=1}^{\infty}$ is not a Cauchy sequence

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Question 34 (JAM 2018 • Q2 • ID: JAM_2018_Q2) MCQ +1 M Neg: −0.33

Q.2 Let $a_n = \frac{b_{n+1}}{b_n}$, where $b_1 = 1$, $b_2 = 1$ and $b_{n+2} = b_n + b_{n+1}$, $n \in \mathbb{N}$. Then $\lim_{n\to\infty} a_n$ is

(A) $\frac{1-\sqrt{5}}{2}$ (B) $\frac{1-\sqrt{3}}{2}$ (C) $\frac{1+\sqrt{3}}{2}$ (D) $\frac{1+\sqrt{5}}{2}$

Question 35 (JAM 2017 • Q59 • ID: JAM_2017_Q59) NAT +2 M Neg: Nil (0)

Q.59 The radius of convergence of the power series

$$\sum_{0}^{\infty} n!\, x^{n^2}$$

is ____________

Question 36 (JAM 2017 • Q55 • ID: JAM_2017_Q55) NAT +2 M Neg: Nil (0)

Q.55 Let $a_n = \sqrt{n}$, $n \geq 1$, and let $s_n = a_1 + a_2 + \cdots + a_n$. Then

$$\lim_{n\to\infty}\left(\frac{a_n/s_n}{-\ln(1-a_n/s_n)}\right) = \_\_\_\_\_\_\_\_$$

Question 37 (JAM 2017 • Q47 • ID: JAM_2017_Q47) NAT +1 M Neg: Nil (0)

Q.47 $$\frac{1}{2\pi}\left(\frac{\pi^3}{1!\,3} - \frac{\pi^5}{3!\,5} + \frac{\pi^7}{5!\,7} - \cdots + \frac{(-1)^{n-1}\pi^{2n+1}}{(2n-1)!\,(2n+1)} + \cdots\right) = \_\_\_\_\_\_\_\_$$

Question 38 (JAM 2017 • Q36 • ID: JAM_2017_Q36) MSQ +2 M Neg: Nil (0)

Q.36 Let $\{x_n\}$ be a real sequence such that $7x_{n+1} = x_n^3 + 6$ for $n \geq 1$. Then which of the following statements is/are TRUE?

(A) If $x_1 = \frac{1}{2}$, then $\{x_n\}$ converges to 1

(B) If $x_1 = \frac{1}{2}$, then $\{x_n\}$ converges to 2

(C) If $x_1 = \frac{3}{2}$, then $\{x_n\}$ converges to 1

(D) If $x_1 = \frac{3}{2}$, then $\{x_n\}$ converges to -3

Question 39 (JAM 2017 • Q28 • ID: JAM_2017_Q28) MCQ +2 M Neg: −0.67

Q.28 Which one of the following is TRUE?

(A) Every sequence that has a convergent subsequence is a Cauchy sequence

(B) Every sequence that has a convergent subsequence is a bounded sequence

(C) The sequence $\{\sin n\}$ has a convergent subsequence

(D) The sequence $\left\{n \cos \frac{1}{n}\right\}$ has a convergent subsequence

Question 40 (JAM 2017 • Q27 • ID: JAM_2017_Q27) MCQ +2 M Neg: −0.67

Q.27

$$\lim_{n\to\infty} \frac{1}{\sqrt{n}}\left(\frac{1}{\sqrt{3}+\sqrt{6}} + \frac{1}{\sqrt{6}+\sqrt{9}} + \cdots + \frac{1}{\sqrt{3n}+\sqrt{3n+3}}\right) =$$

(A) $1+\sqrt{3}$ (B) $\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{1+\sqrt{3}}$

Question 41 (JAM 2017 • Q26 • ID: JAM_2017_Q26) MCQ +2 M Neg: −0.67

Q.26 Let $0 < a_1 < b_1$. For $n \geq 1$, define

$$a_{n+1} = \sqrt{a_n b_n} \quad \text{and} \quad b_{n+1} = \frac{a_n + b_n}{2}.$$

Then which one of the following is NOT TRUE?

(A) Both $\{a_n\}$ and $\{b_n\}$ converge, but the limits are not equal

(B) Both $\{a_n\}$ and $\{b_n\}$ converge and the limits are equal

(C) $\{b_n\}$ is a decreasing sequence

(D) $\{a_n\}$ is an increasing sequence

Question 42 (JAM 2017 • Q21 • ID: JAM_2017_Q21) MCQ +2 M Neg: −0.67

Q.21 The interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{1}{(-3)^{n+2}} \frac{(4x-12)^n}{n^2+1}$$

is

(A) $\frac{10}{4} \leq x < \frac{14}{4}$

(B) $\frac{9}{4} \leq x < \frac{15}{4}$

(C) $\frac{10}{4} \leq x \leq \frac{14}{4}$

(D) $\frac{9}{4} \leq x \leq \frac{15}{4}$

Question 43 (JAM 2017 • Q13 • ID: JAM_2017_Q13) MCQ +2 M Neg: −0.67

Q.13 Let $M = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 0 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Then

$$\lim_{n\to\infty} M^n x$$

(A) does not exist

(B) is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(C) is $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$

(D) is $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$

Question 44 (JAM 2017 • Q9 • ID: JAM_2017_Q9) MCQ +1 M Neg: −0.33

Q.9 If $\lim_{T\to\infty} \int_0^T e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$, then

$$\lim_{T\to\infty} \int_0^T x^2 e^{-x^2}\,dx =$$

(A) $\frac{\sqrt{\pi}}{4}$ (B) $\frac{\sqrt{\pi}}{2}$ (C) $\sqrt{2\pi}$ (D) $2\sqrt{\pi}$

Question 45 (JAM 2017 • Q4 • ID: JAM_2017_Q4) MCQ +1 M Neg: −0.33

Q.4

$$\lim_{n\to\infty} \frac{\pi}{n}\sum_{k=1}^{n} \sin\left(\frac{\pi}{2} + \frac{5\pi}{2}\cdot\frac{k}{n}\right) =$$

(A) $\frac{2\pi}{5}$ (B) $\frac{5}{2}$ (C) $\frac{2}{5}$ (D) $\frac{5\pi}{2}$

Question 46 (JAM 2016 • Q51 • ID: JAM_2016_Q51) NAT +2 M Neg: Nil (0)

Q.51 The value of $\lim_{n\to\infty}\left(8n - \frac{1}{n}\right)^{\frac{(-1)^n}{n^2}}$ is equal to ________

Question 47 (JAM 2016 • Q48 • ID: JAM_2016_Q48) NAT +1 M Neg: Nil (0)

Q.48 The radius of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{(-4)^n}{n(n+1)}(x+2)^{2n} \text{ is } \_\_\_\_\_\_\_\_$$

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Question 48 (JAM 2016 • Q42 • ID: JAM_2016_Q42) NAT +1 M Neg: Nil (0)

Q.42 Let $\{s_k\}$ be a sequence of real numbers, where

$$s_k = k^{\alpha/k}, \qquad k \geq 1, \qquad \alpha > 0.$$

Then

$$\lim_{n\to\infty} \left(s_1\, s_2\, \dots\, s_n\right)^{1/n}$$

is ____________

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Question 49 (JAM 2016 • Q41 • ID: JAM_2016_Q41) NAT +1 M Neg: Nil (0)

Q.41 Let $\{s_n\}$ be a sequence of real numbers given by

$$s_n = 2^{(-1)^n}\left(1 - \frac{1}{n}\right)\sin\frac{n\pi}{2}, \qquad n \in \mathbb{N}.$$

Then the least upper bound of the sequence $\{s_n\}$ is ____________

Question 50 (JAM 2016 • Q31 • ID: JAM_2016_Q31) MSQ +2 M Neg: Nil (0)

Q.31 Let $\{s_n\}$ be a sequence of positive real numbers satisfying

$$2\, s_{n+1} = s_n^2 + \frac{3}{4}, \qquad n \geq 1.$$

If α and β are the roots of the equation $x^2 - 2x + \frac{3}{4} = 0$ and $\alpha < s_1 < \beta$, then which of the following statement(s) is(are) TRUE ?

(A) $\{s_n\}$ is monotonically decreasing
(B) $\{s_n\}$ is monotonically increasing
(C) $\lim_{n\to\infty} s_n = \alpha$
(D) $\lim_{n\to\infty} s_n = \beta$

Question 51 (JAM 2016 • Q24 • ID: JAM_2016_Q24) MCQ +2 M Neg: −0.67

Q.24 The sum of the series

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{n^2+n-2}$$

is

(A) $\frac{1}{3}\ln 2 - \frac{5}{18}$ (B) $\frac{1}{3}\ln 2 - \frac{5}{6}$ (C) $\frac{2}{3}\ln 2 - \frac{5}{18}$ (D) $\frac{2}{3}\ln 2 - \frac{5}{6}$

Question 52 (JAM 2016 • Q12 • ID: JAM_2016_Q12) MCQ +2 M Neg: −0.67

Q.12 Let $\{a_n\}$ be a sequence of positive real numbers satisfying

$$\frac{4}{a_{n+1}} = \frac{3}{a_n} + \frac{a_n^3}{81}, \qquad n \geq 1, \ a_1 = 1.$$

Then all the terms of the sequence lie in

(A) $\left[\frac{1}{2}, \frac{3}{2}\right]$ (B) $[0, 1]$ (C) $[1, 2]$ (D) $[1, 3]$

Question 53 (JAM 2016 • Q11 • ID: JAM_2016_Q11) MCQ +2 M Neg: −0.67

Q.11 Let S be the series

$$\sum_{k=1}^{\infty} \frac{1}{(2k-1)\, 2^{(2k-1)}}$$

and T be the series

$$\sum_{k=2}^{\infty} \left(\frac{3k-4}{3k+2}\right)^{\frac{(k+1)}{3}}$$

of real numbers. Then which one of the following is TRUE?

(A) Both the series S and T are convergent
(B) S is convergent and T is divergent
(C) S is divergent and T is convergent
(D) Both the series S and T are divergent

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Question 54 (JAM 2016 • Q1 • ID: JAM_2016_Q1) MCQ +1 M Neg: −0.33

Q. 1 – Q.10 carry one mark each.

Q.1 The sequence $\{s_n\}$ of real numbers given by

$$s_n = \frac{\sin\frac{\pi}{2}}{1\cdot 2} + \frac{\sin\frac{\pi}{2^2}}{2\cdot 3} + \cdots + \frac{\sin\frac{\pi}{2^n}}{n\cdot(n+1)}$$

is

(A) a divergent sequence
(B) an oscillatory sequence
(C) not a Cauchy sequence
(D) a Cauchy sequence

Question 55 (JAM 2015 • Q57 • ID: JAM_2015_Q57) NAT +2 M Neg: Nil (0)

Q.17

If $\int_0^x \left(e^{-t^2} + \cos t\right)\, dt$ has the power series expansion $\sum_{n=1}^{\infty} a_n x^n$, then a_5, correct upto three decimal places, is equal to ______________

Question 56 (JAM 2015 • Q46 • ID: JAM_2015_Q46) NAT +1 M Neg: Nil (0)

Q.6

If the power series

$$\sum_{n=0}^{\infty} \frac{n!}{n^n} x^{2n}$$

converges for $|x| < c$ and diverges for $|x| > c$, then the value of c, correct upto three decimal places, is ______________

Question 57 (JAM 2015 • Q33 • ID: JAM_2015_Q33) MSQ +2 M Neg: Nil (0)

Q.3

Which of the following conditions implies (imply) the convergence of a sequence $\{x_n\}$ of real numbers?

(A) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $|x_{n+1} - x_n| < \varepsilon$

(B) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $\frac{1}{(n+1)^2}|x_{n+1} - x_n| < \varepsilon$

(C) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $(n+1)^2|x_{n+1} - x_n| < \varepsilon$

(D) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all m, n with $m > n \geq n_0$, $|x_m - x_n| < \varepsilon$

Question 58 (JAM 2015 • Q23 • ID: JAM_2015_Q23) MCQ +2 M Neg: −0.67

Q.23

The sequence $\left\{\cos\left(\frac{1}{2}\tan^{-1}\left(-\frac{n}{2}\right)^n\right)\right\}$ is

(A) monotone and convergent
(B) monotone but not convergent
(C) convergent but not monotone
(D) neither monotone nor convergent

Question 59 (JAM 2015 • Q22 • ID: JAM_2015_Q22) MCQ +2 M Neg: −0.67

Q.22

Which one of the following statements is true for the series $\sum_{n=1}^{\infty}(-1)^n \frac{(2n)!}{n^{2n}}$?

(A) The series converges conditionally but not absolutely
(B) The series converges absolutely
(C) The sequence of partial sums of the series is bounded but not convergent
(D) The sequence of partial sums of the series is unbounded

Question 60 (JAM 2015 • Q5 • ID: JAM_2015_Q5) **MCQ** **+1 M** **Neg: −0.33**

Q.5 Let $\{x_n\}$ be a convergent sequence of real numbers. If $x_1 > \pi + \sqrt{2}$ and $x_{n+1} = \pi + \sqrt{x_n - \pi}$ for $n \geq 1$, then which one of the following is the limit of this sequence?

(A) $\pi + 1$ (B) $\pi + \sqrt{2}$ (C) π (D) $\pi + \sqrt{\pi}$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Sequences and Series of Real Numbers (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	0
2	NAT	+1	1
3	NAT	+1	7
4	MCQ	+2	B
5	MCQ	+2	B
6	NAT	+2	33 to 33
7	NAT	+2	0.25 to 0.25
8	NAT	+1	1 to 1
9	MSQ	+2	A;D
10	MSQ	+2	B;C
11	MSQ	+2	A;D
12	MCQ	+2	D
13	MCQ	+2	A
14	MCQ	+2	D
15	MCQ	+1	D
16	MCQ	+1	C
17	NAT	+2	2.69 to 2.74
18	MSQ	+2	A, B
19	MSQ	+2	A
20	MCQ	+2	A

Q. No.	Type	Marks	Official Key
21	MCQ	+2	C
22	MCQ	+2	B
23	MCQ	+2	A
24	MCQ	+1	A
25	MCQ	+1	D
26	NAT	+2	0.4 to 0.6
27	NAT	+1	2 to 2
28	NAT	+1	1 to 1
29	MCQ	+2	B
30	MCQ	+2	D
31	MCQ	+2	C
32	MCQ	+2	A
33	MCQ	+1	B
34	MCQ	+1	D
35	NAT	+2	0.9 to 1.1
36	NAT	+2	0.9 to 1.1
37	NAT	+1	0.49 to 0.51
38	MSQ	+2	A, C
39	MCQ	+2	C
40	MCQ	+2	C

Q. No.	Type	Marks	Official Key
41	MCQ	+2	A
42	MCQ	+2	D
43	MCQ	+2	C
44	MCQ	+1	A
45	MCQ	+1	C
46	NAT	+2	1.0 to 1.0
47	NAT	+1	0.5 to 0.5
48	NAT	+1	1.0 to 1.0
49	NAT	+1	0.5 to 0.5
50	MSQ	+2	A;C
51	MCQ	+2	C
52	MCQ	+2	D
53	MCQ	+2	B
54	MCQ	+1	D
55	NAT	+2	0.10 to 0.11
56	NAT	+1	1.64 to 1.65
57	MSQ	+2	C;D
58	MCQ	+2	A
59	MCQ	+2	B
60	MCQ	+1	A

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.